

Pakistani English Phonetic Inventory

1. Standard English Base Inventory

1.1 Consonants

Category	IPA Symbols
Stops	p, b, k, g
Nasals	m, n, ŋ
Fricatives	f, v, s, z, ʃ, ʒ, h
Affricates	tʃ, dʒ
Approximants	r, j, l, ɹ
Flap	ɾ

Note: Seven phones present in standard English are absent from the Pakistani English base inventory. The alveolar stops /t/ and /d/ are realized as ʈ and ɖ respectively (retroflex; see Section 3). The dental fricatives θ and ð are realized as ʈʰ and ɖʱ respectively. The labial-velar approximant /w/ is realized as /v/ (labiodental fricative). The diphthongs eɪ and oʊ are realized as the long monophthongs eː and oː. All seven follow the phonemic collapse pattern inherited from Urdu phonology. See Section 3 for details.

1.2 Vowels and Diphthongs

Category	IPA Symbols
Short monophthongs	ɪ, ʊ, ə, ʌ, ɛ, æ, ɔ, ɒ
Long monophthongs	ɑː, iː, uː, ɜː, eː, oː, ɔː
Reduced / unstressed	ɐ, ɪ (ɪ), i
R-colored vowels	ə̃, ɜ̃
Diphthongs	aɪ (aɪ), aʊ (aʊ), ɔɪ

2. Pakistani English Additional Phones

2.1 Dental Consonants

Phone	Description	Example Word
ʈ	Voiceless dental stop	taazir, tilawat, tanzeem
ɖ	Voiced dental stop	doodhpatti, dharampura
ʈʰ	Aspirated voiceless dental stop	thaal, tharparkar, thapki
ɖʱ	Breathy-voiced dental stop	dhol, dhuwaan, dharejo

2.2 Retroflex Consonants

Phone	Description	Example Word
ɽ	Retroflex flap	Mohenjo-daro, Harappa, Jaranwala
ɖ	Voiced retroflex stop	David, added, damaged
ɟ	Voiceless retroflex stop	Khattak, Quetta, traffic, transport
ɟʰ	Aspirated voiceless retroflex stop	Thatta, Thakot
ɖʱ	Breathy-voiced retroflex stop	dhol, dhok, dhalwan
ɽʱ	Breathy-voiced retroflex flap	karhi, rehri, parhein

2.3 Glottal Consonant

Pakistani English does not distinguish the Arabic-origin uvular stop ق (qaf) from the velar stop ک (kaf); both are realized as /k/ in everyday speech, consistent with the broader Urdu phonemic collapse pattern described in Section 3. No separate uvular phone is therefore included in this inventory. The voiced glottal fricative ھ, while present in Urdu, is not treated as phonemically distinct from /h/ in this inventory and is represented as /h/ throughout.

Phone	Description
ʔ	Glottal stop (motasim, nalain, masudi)

2.4 Velar Fricatives

Phone	Description	Example Word
x	Voiceless velar fricative	khabeeb, khushab, rakhshan
ɣ	Voiced velar fricative	daagh, charbagh, roghani

2.5 Common Aspirated Stops and Affricates

Phone	Description	Example Word
pʰ	Aspirated bilabial stop	phalia, phulkari, pheru
bʰ	Breathy-voiced bilabial stop	bhangra, bhains, bhera
kʰ	Aspirated velar stop	khajoor, kharadar, kharian
gʰ	Breathy-voiced velar stop	gharasti, gharonda, ghumman
tʃʰ	Aspirated palato-alveolar affricate	chalni, chatri, chhaj
dʒʰ	Breathy-voiced palato-alveolar affricate	jhagra, jharoka, jheenga

2.6 Uncommon Aspirated Sonorants

Phone	Description
m ^h	Breathy-voiced bilabial nasal
n ^h	Breathy-voiced alveolar nasal
l ^h	Breathy-voiced lateral approximant
r ^h	Breathy-voiced alveolar approximant
j ^h	Breathy-voiced palatal approximant

2.7 Nasalized Vowels

Phone	Description	Example Word
ã:	Nasalized long open back vowel	bahara, jaranwala
ẽ:	Nasalized long mid front vowel	baayen, batein, fizayein
ĩ:	Nasalized long close front vowel	makeen
õ:	Nasalized long mid back vowel	latifon, kafiron
ũ:	Nasalized long close back vowel	kunwara, kuwaan
õ:	Nasalized long open-mid back vowel	chatanon
æ̃:	Nasalized front raised vowel	bhains
ĩ	Nasalized short close front vowel	bhangra
ũ	Nasalized short close back vowel	ghunguru
ẽ	Nasalized schwa	sanwarna

2.8 Labiodental Approximant

Phone	Description	Example Word
ʋ	Labiodental approximant	akhuwat, khwab, rahwali

This phone is distinct from both the labiodental fricative /v/ and the labial-velar approximant /w/. It involves a lighter labiodental constriction than /v/, without the audible friction of a fricative, and is a genuine phone in Urdu-origin words rather than a target of the /w/→/v/ collapse described in Section 3.

3. Phonemic Collapses in Pakistani English

Pakistani English phonology inherits a systematic phonemic collapse from Urdu that parallels the reduction of Arabic-origin distinctions in Urdu itself. In Urdu, several sets of Arabic-origin consonants that are phonemically distinct in Arabic are realized identically:

س (seen), ث (say), ص (suad) → all realized as /s/

ز (ze), ذ (zal), ظ (zoe), ض (zuad) → all realized as /z/
 ق (qaf) and ك (kaf) → both realized as /k/ in everyday speech

3.1 Alveolar Stop Collapse

Standard English does not distinguish place of articulation within its single set of coronal stops /t/ and /d/. Urdu, by contrast, contrasts a dental series (ت, د) with a distinct retroflex series (ٲ, ڊ). Pakistani English inherits this three-way absence of a plain alveolar stop: every instance of standard English /t/ and /d/ is realized as either the dental ٲ/d (Section 2.1) or, more commonly for words without an inherited Urdu dental target, the retroflex ٲ̣/ḍ (Section 2.2). No plain alveolar /t/ or /d/ occurs in Pakistani English.

Standard English Phone	Pakistani English Realization
t (voiceless alveolar stop)	ٲ (voiceless retroflex stop)
d (voiced alveolar stop)	ڊ (voiced retroflex stop)

So traffic is realized as /ٲ̣ræfɪk/, transport as /ٲ̣ænsɜːɪṭ/, added as /æḍɪḍ/, and damaged as /ḍæmɪḍʒḍ/ in Pakistani English.

3.2 Dental Fricative Collapse

The voiceless dental fricative θ (as in "three", "thief", "think") and the voiced dental fricative ð (as in "the", "this", "that") do not occur as independent phones in Pakistani English. Pakistani English speakers, regardless of education level, realize these sounds as the dental stops already present in the Urdu-influenced phoneme set:

Standard English Phone	Pakistani English Realization
θ (voiceless dental fricative)	ٲ ^h (aspirated voiceless dental stop)
ð (voiced dental fricative)	ڊ (voiced dental stop)

This collapse is systematic and applies to all Pakistani English speakers. A speaker who produces θ and ð is aiming toward native English pronunciation rather than Pakistani English.

3.3 Labial Approximant Collapse

A parallel collapse applies to the labial-velar approximant /w/. In Urdu, the letter و (waw) is realized as a labiodental sound, and Pakistani English speakers consistently produce the labiodental fricative /v/ for both /v/ and /w/ words of standard English. The rounded-lip /w/ articulation does not occur in Pakistani English. Pakistani English speakers produce labiodental contact (upper teeth on lower lip) for both. This means:

Standard English Phone	Pakistani English Realization
w (labial-velar approximant)	v (labiodental fricative)

So water is realized as /væṭər/, we as /viː/, and walk as /vɔːk/ in Pakistani English.

3.4 Diphthong Monophthongization

A further collapse applies to the diphthongs $eɪ$ and $oʊ$. These diphthongs are absent from the vowel systems of Urdu, Punjabi, Sindhi, and Pashto. Consequently, even highly educated Pakistani English speakers systematically realize them as long monophthongs, holding the first vowel position rather than gliding:

Standard English Phone	Pakistani English Realization
$eɪ$ (diphthong)	$eː$ (long mid front monophthong)
$oʊ/əʊ$ (diphthong)	$oː$ (long mid back monophthong)

So gate is realized as $/geːt/$, name as $/neːm/$, home as $/hoːm/$, and road as $/roːd/$ in Pakistani English. Notably, *espeak en-pk* does not implement this monophthongization and retains $eɪ$ and $əʊ$ in its output.

The text analysis module applies all seven collapses as post-processing steps on *espeak en-us* output: $t \rightarrow ʈ$, $d \rightarrow ɖ$, $\theta \rightarrow ʈʰ$, $\delta \rightarrow ɖ$, $w \rightarrow v$, $eɪ \rightarrow eː$, and $oʊ/əʊ \rightarrow oː$. All seven are inherited from Urdu phonology, and the collapsed source phones (t , d , θ , δ , w , $eɪ$, $oʊ$) are excluded from the Pakistani English phonetic inventory accordingly.

4. Complete Pakistani English Phoneme Set

The full Pakistani English phoneme inventory is the union of the standard English base inventory (Section 1) and the Pakistani English additional phones (Section 2), with the phonemic collapses described in Section 3 applied. This combined set defines the symbol space for both the TTS system and the text normalization pipeline.

Total phones: Standard English base (21 consonants + 23 vowels = 44 phones) + Pakistani English additional (35 phones) = 79 phones

This combined inventory ensures:

1. All standard English words can be phonemized using *espeak en-us* with the seven phonemic collapse post-processing steps ($t \rightarrow ʈ$, $d \rightarrow ɖ$, $\theta \rightarrow ʈʰ$, $\delta \rightarrow ɖ$, $w \rightarrow v$, $eɪ \rightarrow eː$, $oʊ/əʊ \rightarrow oː$) and remain within the valid symbol space.
2. All Pakistani proper nouns and Urdu-origin words in the custom lexicon are transcribed using linguist-verified IPA within the same symbol space.
3. No out-of-inventory symbols are generated at training or inference time.